Xu Gao

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EDUCATION

Undergraduate, Xingjian College, Tsinghua University **Sep.2021-Jun.2026**

Double Major: Theoretical and Applied Mechanics & Energy and Power Engineering
Certificate: Industrial Engineering and Data Science, Department of Industrial Engineering & Artificial Intelligence Innovation and Entrepreneurship Enhancement Program, Fundamental Industry Training Center

RESEARCH EXPERIENCE

Artifact-Grounded Benchmark for Autonomous DFT Agent Frameworks **Jul.2026-Present**

- Researcher in Yuyu Luo (骆昱宇)'s lab, DSA, HKUST-GZ.
  - **Workflow-Level Benchmarking**: Developing a benchmark framework to evaluate autonomous DFT agent frameworks at the scientific workflow level, moving beyond isolated tool-use tasks and final-output correctness to assess end-to-end scientific decision-making.
  - **Artifact-Grounded Evaluation**: Establishing evidence-based criteria for convergence verification, computational provenance, and downstream workflow compatibility, with controlled counterfactual tests to examine whether workflow transitions and computational-reuse decisions are justified by scientific evidence.

AgentLab for Perovskite Materials: An Agentic Platform for Scientific Workflows **Apr.2026-Present**

- Researcher in Yuyu Luo (骆昱宇)'s lab, DSA, HKUST-GZ.
  - **Platform Architecture**: Designed an extensible agent platform with modular MCP-based interfaces for orchestrating scientific software and simulation workflows, with a pathway toward laboratory-instrument integration.
  - **Scientific Workflow Integration**: Built a workflow connecting natural-language task specification, HPC/SLURM job orchestration, VASP-based first-principles calculations, automated result parsing, and SCAPS device simulation, enabling structured data transfer across scientific tools.

Photovoltaic Power Generation Ultra-Short-Term Forecasting by ML & CV **Jan.2025-Jun.2025**

- Researcher in Shuiqing Li (李水清) 's lab, Department of Energy and Power Engineering, THU.
  - Developed a novel hybrid model integrating cloud motion and shadow generation for PV power prediction.
  - Awarded Outstanding Undergraduate Thesis in Energy & Power Engineering, Tsinghua University.

Downscaling of Long-Range Climate Prediction Models **Nov.2024-Apr.2025**

- Researcher in Da Zhang (张达) 's lab, School of Environment, THU.
  - Adapted NREL Super3 model for high-resolution climate downscaling over China (Anhui case study).
  - Processed 18 years of data, achieving 30km/24hr resolution (vs. traditional 100km/72hr).

Structural engineering of Pt-based intermetallic catalysts for Fuel Cells **Feb.2024-Jun.2024**

- Researcher in Liang Zhang (张亮) 's lab, Center for Combustion Energy, THU.
  - Designed PtCoMn/PtCoTi trimetallic catalysts using DFT calculations and Monte Carlo simulations.
  - Validation through HAADF-STEM characterization and electrochemical tests.

Cage-shaped Self-folding Mechanical Metamaterials **Sep.2022-May.2023**

- Co-Authorship, researcher in Changqing Chen (陈常青) 's lab, Department of Aerospace Engineering, THU.
  - Validation through theories, Abaqus and other relevant techniques.
  - Experiments: set up the experiment environment, do lab design, and data simulation, etc.
  - Writing part of the paper and editing & reviewing, published in IJSS.

TECHNICAL SKILLS

Programming: Python, Matlab
Scientific Computing: VASP, SCAPS, HPC/SLURM
Modeling: Machine Learning, Computer Vision, Monte Carlo, Abaqus, COMSOL, SolidWorks

PUBLICATION

- Meng, Z., Gao, X., Yan, H., Liu, M., Cao, H., Mei, T., & Qing Chen, C. (2024). Cage-shaped self-folding mechanical metamaterials. International Journal of Solids and Structures, 286–287, 112560. https://doi.org/10.1016/j.ijsolstr.2023.112560

HONORS & AWARDS

- Grand Prize of the 41st Challenge Cup of Tsinghua University, First Author
- Outstanding Undergraduate Thesis in Energy and Power Engineering, Tsinghua University, 2025